How LLMs Actually Works

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Homework: Find three tasks where the AI gives you a wrong answer. Write down why you think it failed.

A Large Language Model (LLM) is a type of deep learning trained on vast amounts of text data to understand, summarize, generate, and predict new content.

Task1:

Gemini model:

I asked google Gemini: Generate a clear, high quality picture of a 12year boy helping his grandmother in the kitchen.

Gemini generated a perfect picture so I said Perfect. Then Gemini didn't understand and generated the same picture again and then I corrected the prompt "I have given you compliment "Perfect" about the 1st picture but you sent another picture. Then I asked why did you send picture? Then you again sent. Please stop sending picture.

Reason of Failure:

I think I have given an incomplete prompt.

And Gemini is restricted with text-only explanations without calling the generation tool.

Task 2:

ChatGPT:

I asked ChatGPT:

"Write a summary of chapter no. 2 of physics book."

Here model gave output successfully. Asking about class/grade and Board and gave summary.

Reason of Success:

Because it is programmed to gather context to tailor its response, and its training data contains billions of educational texts and textbooks.

Then:

I asked: "Give me summary of 3rd chapter of English book of class 3 of Sindh Text Book.

Model generated a summary.

Reason of Failure:

Hallucination. Because no such chapter exists in English book also model didn't ask me about author or publisher.

Task 3:

I gave prompt: “write code for login form”

It gave me the long code without asking about the language and other details.

Reason of Failure:

Incomplete Prompt like missing context may be the reason.